DRAFT for DICTO to read — not published. The exact proof of the halves is still to be done (marked below). Drafted by the local model on dicto-big-brother, corrected and checked by Claude against the study.

Pyrochlore Halves

The rule of halves of #57 carried to the pyrochlore network, its layers stacked in any order

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The idea is DICTO's. Computation with Claude (Anthropic).

In short. The pyrochlore network is made of small tetrahedra that touch only at their corners. It hides inside every diamond-like crystal: take the middle of every bond, and the four middles round each atom make one of these tetrahedra. Diamond-like crystals can stack their layers in many orders, so the pyrochlore network can too. We share out all of space round it by one weight, as in #57, and find the same rule: every piece is borrowed, whole or in half, from the pure cubic or the pure hexagonal stacking.

1. The idea

DICTO's idea (27 October 2026): "close packing halves rule pyrochlore network" — the rule of halves of #57 (DICTO DOI 10.5281/zenodo.23647898) on the pyrochlore network, its layers (kagome and triangular, seen down a three-fold axis) stacked in any order. The series paper #58 (DICTO DOI 10.5281/zenodo.23665944) tells the road to #57.

2. Main result

Rule (proved exactly for the walks; for the halves, checked and to be proved). In every close-packed stacking, divide space by weight among the balls and diamond atoms (weight λ) and the two empty holes of each gap (weight 0). Then every piece is a piece of a pure stacking: a ball's piece is the ball piece of its own layer's pure stacking (cubic if the layer is c, hexagonal if h); a diamond atom's piece that of its nearest layer; and each hole piece, cut at its gap's mid-height, is a bottom half from the layer below and a top half from the layer above.

> **Status:** checked on the stackings 4H, 6H, 9R and 15R at $\lambda = -1/4, -1/5, -1/10, 0$ and $1/10$: 204 pieces at each weight, every one equal to its pure piece (float, 7 decimals). [CHECK: the exact proof, by #57's method — locality over the 16 cubic/hexagonal patterns of six layers, then exact corner functions along each stretch of the weight.]

3. Construction

Balls of spacing 1 in close-packed layers $\sqrt{(2/3)}$ apart, stacked by a word in A, B, C. The diamond-type crystal (as silicon carbide) is the balls plus #57's tetrahedral hole "over X at 3/4" above each ball. The middles of its bonds form the pyrochlore network: round each atom four middles make a regular tetrahedron of edge 1/2, and every corner belongs to exactly two tetrahedra (checked for 3C, 2H, 4H and 6H). Each gap holds four sites: the ball S, the

diamond atom C, the octahedral hole O and the other tetrahedral hole Y; the two empty holes are exactly the network's open spaces.

4. The weight walk

Computed exactly with #57's census engine, unchanged (rational arithmetic; every built-in check passed).

Cubic (3C). Stops at −3/8 (the atoms' pieces vanish), **−1/4** (atoms regular tetrahedra, holes truncated tetrahedra of volume $23\sqrt{2}/96$: the classical tetrahedral–truncated tetrahedral honeycomb), **+1/4** (the roles swap: atoms truncated tetrahedra, holes tetrahedra), and +3/8 (the holes' pieces vanish).

Hexagonal (2H). Stops at −11/36, −1/4, −3/20, 0, 1/8, 1/6 and 1/4. At −1/4 the diamond atoms' pieces are still regular tetrahedra but the balls' pieces have ten corners: the regular pyrochlore tetrahedra are exact pieces only on cubic layers.

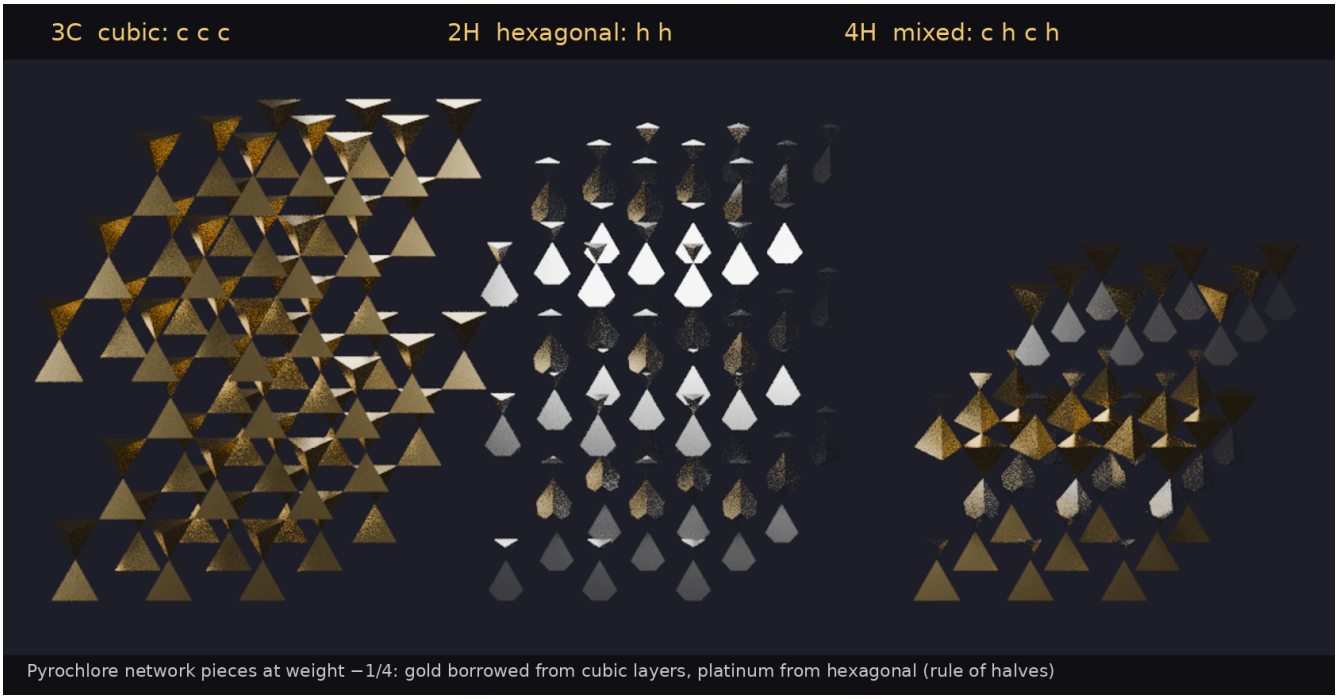


Figure 1. The network's pieces at λ = −1/4 in the cubic (gold), hexagonal (platinum) and mixed 4H stackings; open spaces left out.

5. Prior art and credit

The pyrochlore lattice as the bond-middle lattice of diamond, and as alternating kagome and triangular layers along a three-fold axis, is standard in the physics of frustrated magnets; the cubic and hexagonal stackings of diamond-type crystals (diamond and lonsdaleite, zincblende and wurtzite) are standard; the tetrahedral–truncated tetrahedral honeycomb is classical; the rule of halves is #57 (DICTO). Not found in the sources searched (a library of 302 references, and the web, 27 October 2026): the rule of halves for pyrochlore stackings, or this weighted division of them.

6. Methods and data

Study folder k-research studies/2026-10-27-pyrochlore-halves: pyro.py (construction), cells.py and halves.py (float division and halves), pyro_walk.py (exact walks with #57's engine), figures.

7. Name

Placeholder until DICTO chooses: Pyrochlore Halves.